

Cart Abandonment Analysis Report

Task #151 - Mankind Matrix Platform

Prepared By:	Khyati Chauhan
Date:	April 2026
Project:	Mankind Matrix - AI & Semiconductor E-commerce Platform

Executive Summary

This report analyzes shopping cart behavior on the Mankind Matrix platform to understand why customers abandon their carts before completing purchases. By examining cart data from August 2025, we identified key patterns that reveal opportunities to recover lost revenue and improve the customer experience.

Key Findings

Metric	Value
Cart Abandonment Rate	45%
Lost Revenue	\$18,024
Conversion Rate	40%
Total Carts Analyzed	100

Cart Status Breakdown

Our analysis of 100 shopping carts revealed the following distribution:

Status	Count	Percentage
Abandoned	45	45%
Converted	40	40%
Active	10	10%
Removed	5	5%

Most Abandoned Products

High-value products are being left behind most often. This suggests customers may be experiencing price hesitation or seeking better deals elsewhere. The iPhone 16 Pro alone accounts for \$3,597 in lost revenue.

Product	Times Left	Lost Revenue
iPhone 16 Pro	3	\$3,597
ThunderCore AI GPU	2	\$2,598
SpectraForce X Series	2	\$1,598
iPhone 16	1	\$999

Key Insights

- 1. High-Value Cart Abandonment:** Premium products like iPhone 16 Pro and ThunderCore AI GPU are frequently abandoned. Customers add these items but hesitate at checkout, possibly due to price sensitivity or comparison shopping.
- 2. Guest User Behavior:** A significant portion of abandoned carts belong to guest sessions. These users may not feel committed without an account, making them more likely to leave without purchasing.
- 3. Lost Revenue Impact:** With \$18,024 sitting in abandoned carts, even recovering 20% through targeted interventions would add \$3,600 in revenue.
- 4. Conversion Opportunity:** The 10% of active carts represent immediate conversion opportunities. These customers are still engaged and can be nudged to complete their purchase.

Recommendations

- 1. Implement Cart Recovery Emails:** Send automated reminder emails to registered users within 24 hours of cart abandonment. Include product images and a clear call-to-action.
- 2. Offer Limited-Time Discounts:** For high-value abandoned items like iPhones and GPUs, consider offering a 5-10% discount or free shipping to incentivize completion.
- 3. Simplify Guest Checkout:** Reduce friction for guest users by minimizing required fields and offering express checkout options.
- 4. Add Exit-Intent Popups:** When users move to leave the checkout page, display a popup offering assistance or a small incentive to complete the purchase.

Technical Implementation

SQL Views Created:

- cart_abandonment_summary: Aggregates cart counts and values by status
- cart_abandonment_rate: Calculates overall abandonment rate and lost revenue
- cart_trend_daily: Tracks daily cart creation and conversion patterns
- abandoned_products: Identifies which products are most frequently abandoned

Data Sources:

- cart: Shopping cart records with status tracking
- cart_item: Individual products within each cart